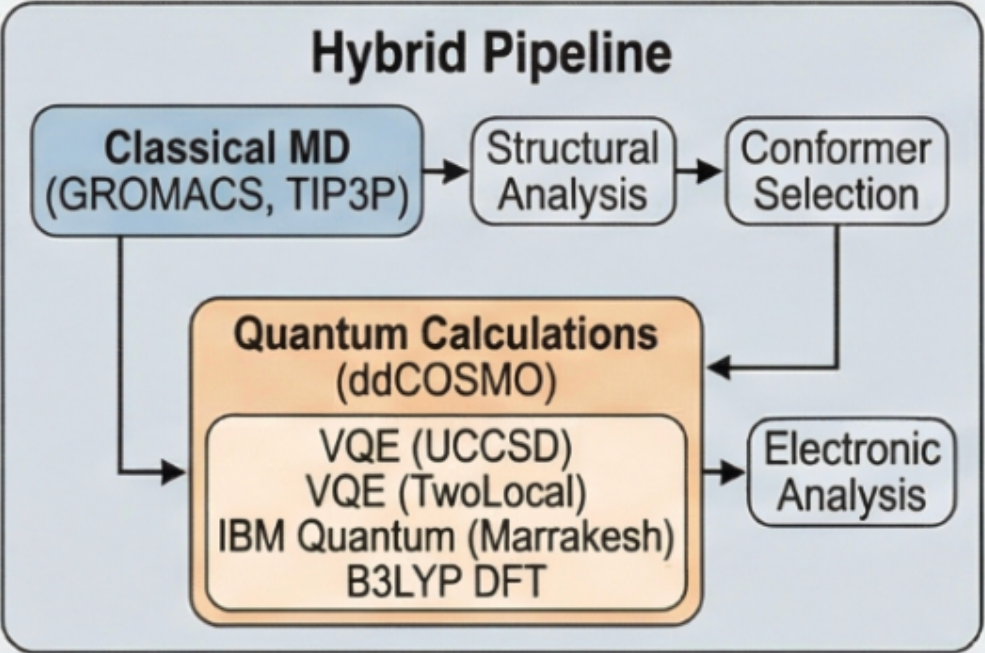
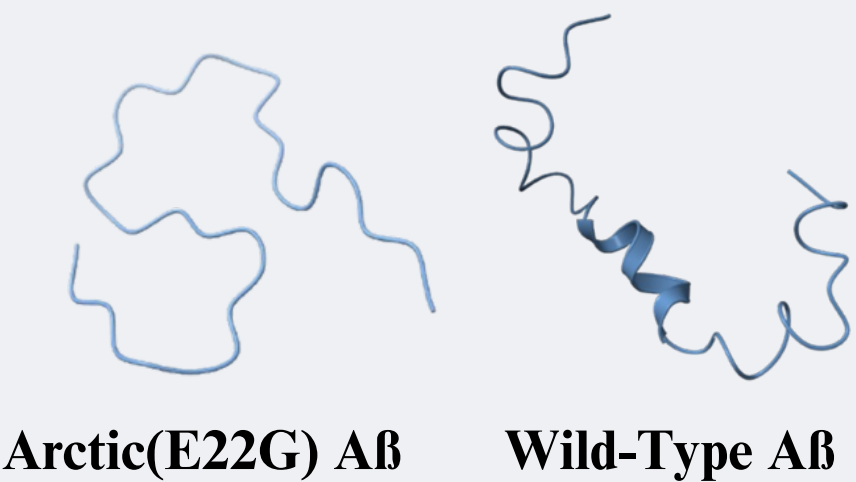


A Hybrid Quantum-Classical Computational Study of Energetic and Conformational Differences in Amyloid-beta variants

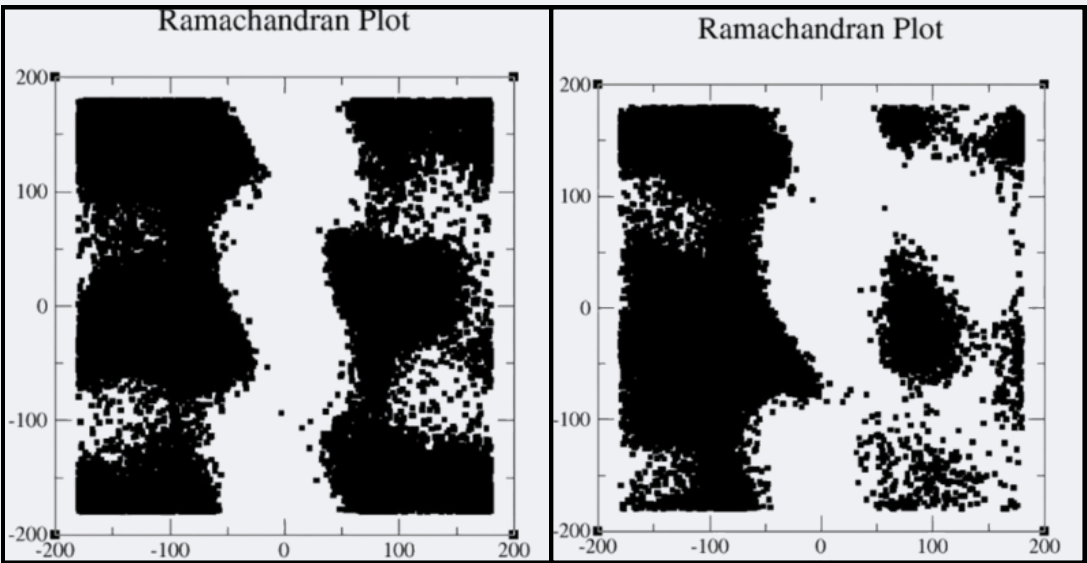
For Table of Contents Only

A. Computational Pipeline & Problem



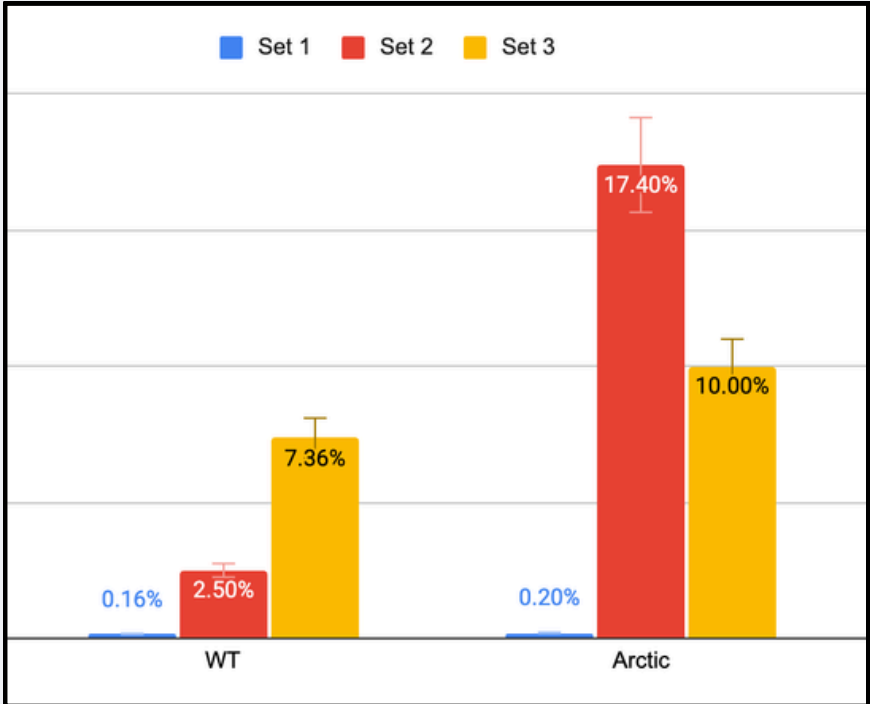
Goal: Elucidate electronic role in accelerated aggregation

B. Classical Structural Reorganization



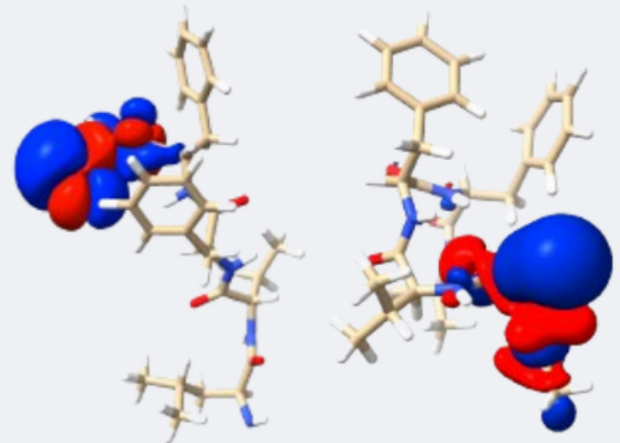
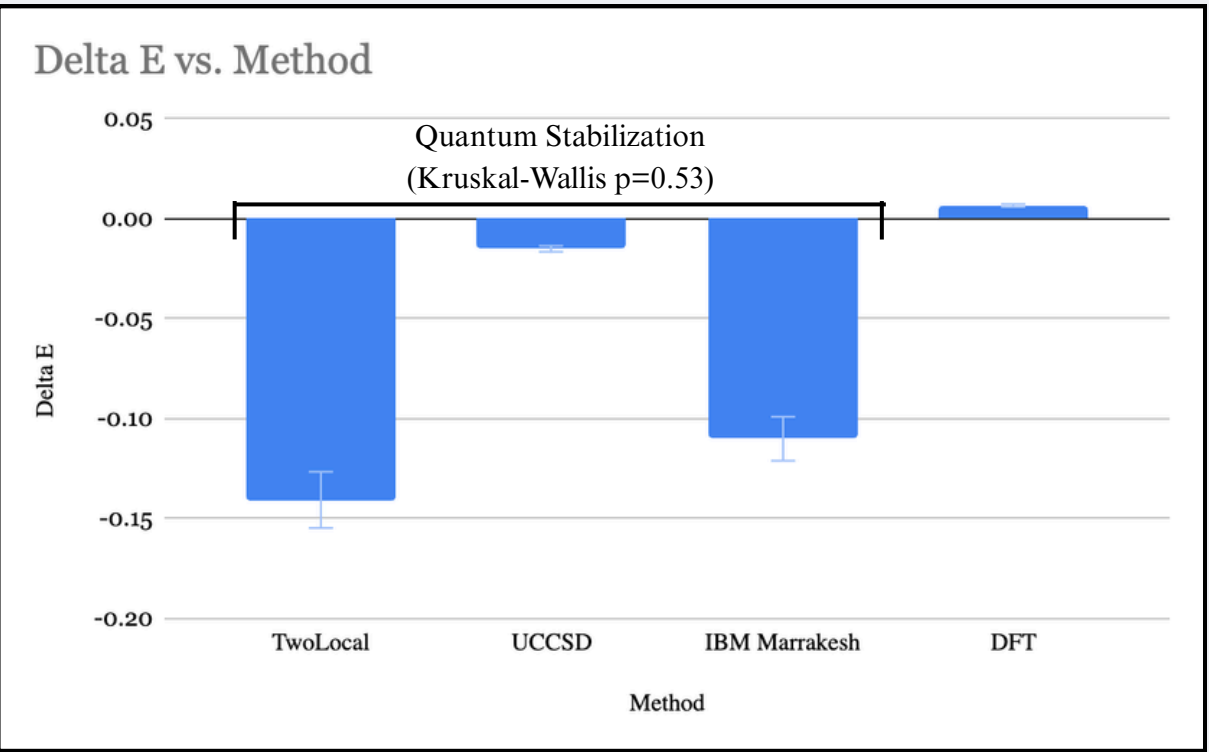
Densely occupied β -favorable angles- Arctic(E22G) Less sparsely occupied β -favorable angles- WT

Kuiper Test $p < 0.001$
(Φ, Ψ)



Arctic favors β -sheet structure

C. Quantum Electronic Stabilization



Arctic(E22G), Set 2, Orbital 150 WT, Set 2, Orbital 150

Arctic mutation confines electron density

D. Integrated Mechanism & Implications

